

VERSION 1.3

JANUARY 2026



DATA STRUCTURE

UAP

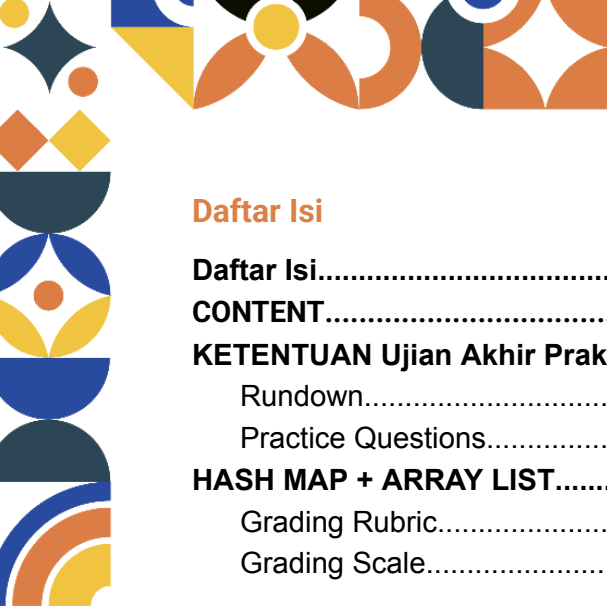
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A vertical strip of colorful geometric patterns, including stars, circles, and abstract shapes in blue, orange, and yellow, runs down the left side of the page.

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CONTENT

KETENTUAN Ujian Akhir Praktikum (UAP)

MOHON DIBACA 3X DENGAN SEKSAMA !!!

1. Seluruh HP wajib dikumpulkan di tengah atau diamankan asisten masing-masing dan dipastikan dalam mode silent tanpa vibrate sebelum ujian dimulai.
2. Soal hanya akan dibagikan setelah semua HP dikumpulkan dan laptop telah diperiksa serta dipastikan aman untuk menghindari kecurangan.
3. Pastikan laptop tanpa akses internet dan mode pesawat aktif. Kemudian untuk password laptop di non - aktifkan.
4. **Tidak boleh buka modul** Praktikan diperbolehkan membuka resume tulis tangan (1 folio bergaris 1 lembar bolak-balik)
5. Waktu pengerjaan UAP adalah 45 menit.
6. Setelah selesai, praktikan keluar dari ruangan, namun belum diperbolehkan mengambil HP.
7. HP hanya boleh diambil setelah seluruh proses penilaian selesai.
8. Live Coding.
9. Silahkan dipelajari modul 2-6, nanti setiap mahasiswa akan mendapatkan soal acak.
10. Tulis nama dan Nim di awal baris kode
11. **Dipastikan laptop aman dan siap untuk UAP, tidak menerima alasan apapun, seperti laptop rusak, baterai habis, charger ketinggalan, dll.**
12. Jika dirasa terdapat kendala segera mencari solusi untuk meminjam laptop teman dan solusi sejenis.
13. Tidak ada susulan UAP.

Rundown

Durasi	Kegiatan
15 Menit	Persiapan & Masuk Ruangan
45 menit	Sesi Pengerjaan
40 Menit	Koreksi & Sterilisasi Ruangan



Practice Questions

HASH MAP + ARRAY LIST

Story

The Bendahara of Majapahit Kingdom records all market transactions. Each transaction contains a trader name and item quantity sold. The system must total sales per trader, then print a ranked leaderboard.

Code (Skeleton)

```
import java.util.*;

public class SoalHashMap {

    // SECTION 1:
    public static Map<String, Integer> countSales(String[] transactions) {
    }

    // SECTION 2: SORT DESCENDING
    public static void printRanking(Map<String, Integer> salesMap) {

    }

    public static void main(String[] args) {
        String[] transactions = {
            "Arya:30","Bima:50","Arya:20","Candra:40",
            "Bima:10","Candra:60","Arya:15"
        };
    }
}
```

Input Data (Hardcoded)

Transactions:{"Arya:30","Bima:50","Arya:20","Candra:40","Bima:10","Candra:60","Arya:15"}

Output Format

```
=== Majapahit Trade Market ===
#1  Candra  : 100 items  [TOP TRADER]
#2  Arya    : 65 items
#3  Bima    : 60 items
Total transactions: 7
```



Key Answer

```
import java.util.*;

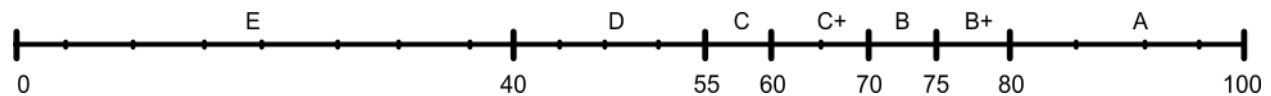
public class SoalHashMap_Answer {
    // * SECTION 1
    public static Map<String, Integer> countSales(String[] transactions) {
        Map<String, Integer> map = new HashMap<>();
        for (String t : transactions) {
            String[] parts = t.split(":");
            String name = parts[0];
            int qty = Integer.parseInt(parts[1]);
            if (map.containsKey(name)) map.put(name, map.get(name) + qty);
            else map.put(name, qty);
        }
        return map;
    }
    // * SECTION 2
    public static void printRanking(Map<String, Integer> salesMap) {
        ArrayList<Map.Entry<String, Integer>> list =
            new ArrayList<>(salesMap.entrySet());
        list.sort((a, b) -> b.getValue() - a.getValue());
        for (int i = 0; i < list.size(); i++) {
            String medal = (i == 0) ? " [TOP TRADER]" : "";
            System.out.println(" " + (i+1) + " " + list.get(i).getKey() + " : " +
list.get(i).getValue() + " items" + medal);
        }
    }
    // * KESESUAIAN IMPLEMENTASI - main
    public static void main(String[] args) {
        String[] transactions = {
            "Arya:30", "Bima:50", "Arya:20", "Candra:40",
            "Bima:10", "Candra:60", "Arya:15"
        };
        System.out.println("=== Majapahit Trade Market ===");
        Map<String, Integer> sales = countSales(transactions);
        printRanking(sales);
        System.out.println("Total transactions: " + transactions.length);
    }
}
```



Grading Rubric

Assessment Aspect	Points
Code	Total 100%
Menerapkan Section 1	35%
Menerapkan Section 2	35%
Kesesuaian implementasi	15%
Menjelaskan program dan kode berjalan	15%

Grading Scale



- A** = (81 - 100) → Sepuh
- B+** = (75 - 80) → Very Good
- B** = (70 - 74) → Good
- C+** = (60 - 69) → Fairly Good
- C** = (55 - 59) → Fair
- D** = (41 - 54) → Poor
- E** = (0 - 40) → Bro really...

